

Unit 9 Pre-Test

Volume of Cylinders, Cones & Spheres

Standard assessed: NC.8.G.9

Name: _____ Date: _____ Period: _____

SCORE 3.0 Basic Understanding

NC.8.G.9 Find the volume of a cylinder and a cone from the formula.

• I can substitute into $V = \pi r^2 h$ and $V = \frac{1}{3}\pi r^2 h$ to find a volume.

Match each vocabulary term to its definition, then find each volume.

A ▾ radius ✓	A. The distance from the center of a circle to its edge.
D ▾ diameter ✓	B. The amount of space a solid figure holds.
C ▾ cylinder ✓	C. A solid with two parallel circular bases.
E ▾ cone ✓	D. The distance across a circle through its center.
F ▾ sphere ✓	E. A solid with one circular base that narrows to a point.
B ▾ volume ✓	F. A perfectly round solid, like a ball.

Volume formulas — reference

Cylinder: $V = \pi r^2 h$ · Cone: $V = \frac{1}{3}\pi r^2 h$ · Sphere: $V = \frac{4}{3}\pi r^3$

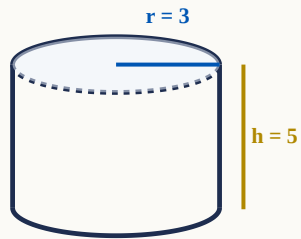
Leave answers in terms of π (for example, 12π) unless told to round.

1. Cylinder

$r = 3$, $h = 5$. Find the volume (in terms of π).

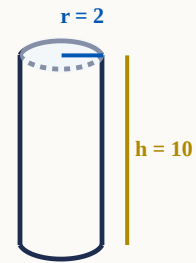
2. Cylinder

$r = 2$, $h = 10$. Find the volume.



$V =$

answer: 45π

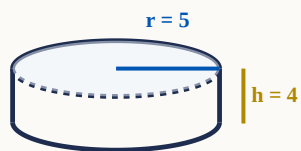


$V =$

answer: 40π

3. Cylinder

$r = 5$, $h = 4$. Find the volume.

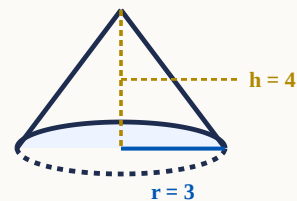


$V =$

answer: 100π

4. Cone

$r = 3$, $h = 4$. Find the volume.

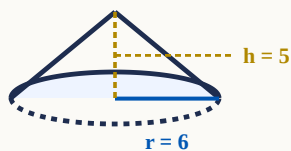


$V =$

answer: 12π

5. Cone

$r = 6$, $h = 5$. Find the volume.

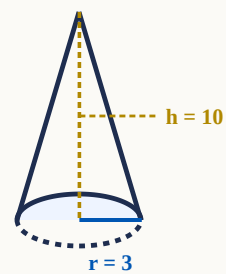


$V =$

answer: 60π

6. Cone

$r = 3$, $h = 10$. Find the volume.



$V =$

answer: 30π

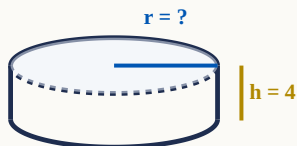
NC.8.G.9 Work backwards from a known volume to find a missing dimension.

• I can rearrange the volume formula to find a missing radius or height.

The volume is given. Find the missing measurement.

7. Find the radius

A cylinder has volume 100π and height 4. Find the radius.

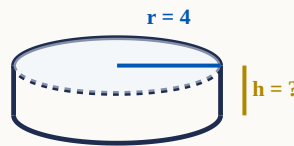


$r =$

answer: 5

8. Find the height

A cylinder has volume 48π and radius 4. Find the height.

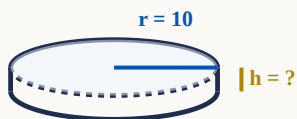


$h =$

answer: 3

9. Find the height

A cylinder has volume 200π and radius 10. Find the height.

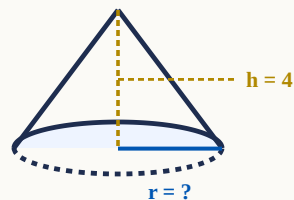


$h =$

answer: 2

10. Find the radius

A cone has volume 12π and height 4. Find the radius.

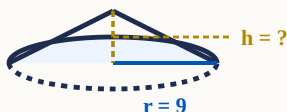


$r =$

answer: 3

11. Find the height

A cone has volume 54π and radius 9. Find the height.

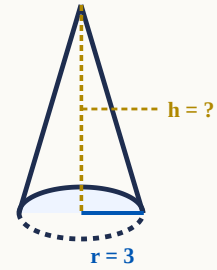


$h =$

12. Find the height

A cone has volume 30π and radius 3. Find the height.

answer: 2



$h =$

answer: 10

SCORE **5.0** Advanced Content

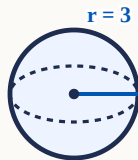
NC.8.G.9 Find the volume of a sphere and solve multi-step problems.

- I can find the volume of a sphere and compare volumes of different solids.

Solve each problem. Leave answers in terms of π unless told otherwise.

13. Sphere

$r = 3$. Find the volume.

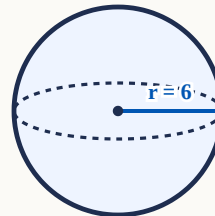


$V =$

answer: 36π

14. Sphere

$r = 6$. Find the volume.

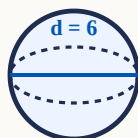


$V =$

answer: 288π

15. Sphere

A sphere has diameter 6. Find the volume.

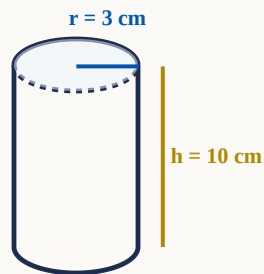


16. Apply

A can (cylinder) has $r = 3$ cm and $h = 10$ cm. Find the volume in terms of π .

V =

answer: 36π

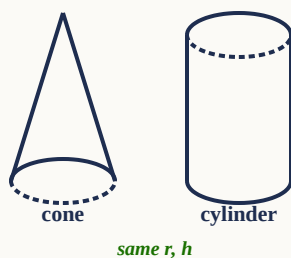


V =

answer: $90\pi \text{ cm}^3$

17. Compare

A cone and a cylinder have the same radius and height. The cone's volume is what fraction of the cylinder's?

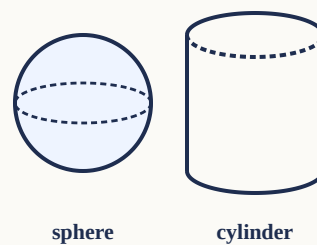


Answer:

answer: $\frac{1}{3}$

18. Compare

A sphere has $r = 3$. A cylinder has $r = 3$, $h = 6$. Which has the greater volume?



Answer:

answer: the cylinder